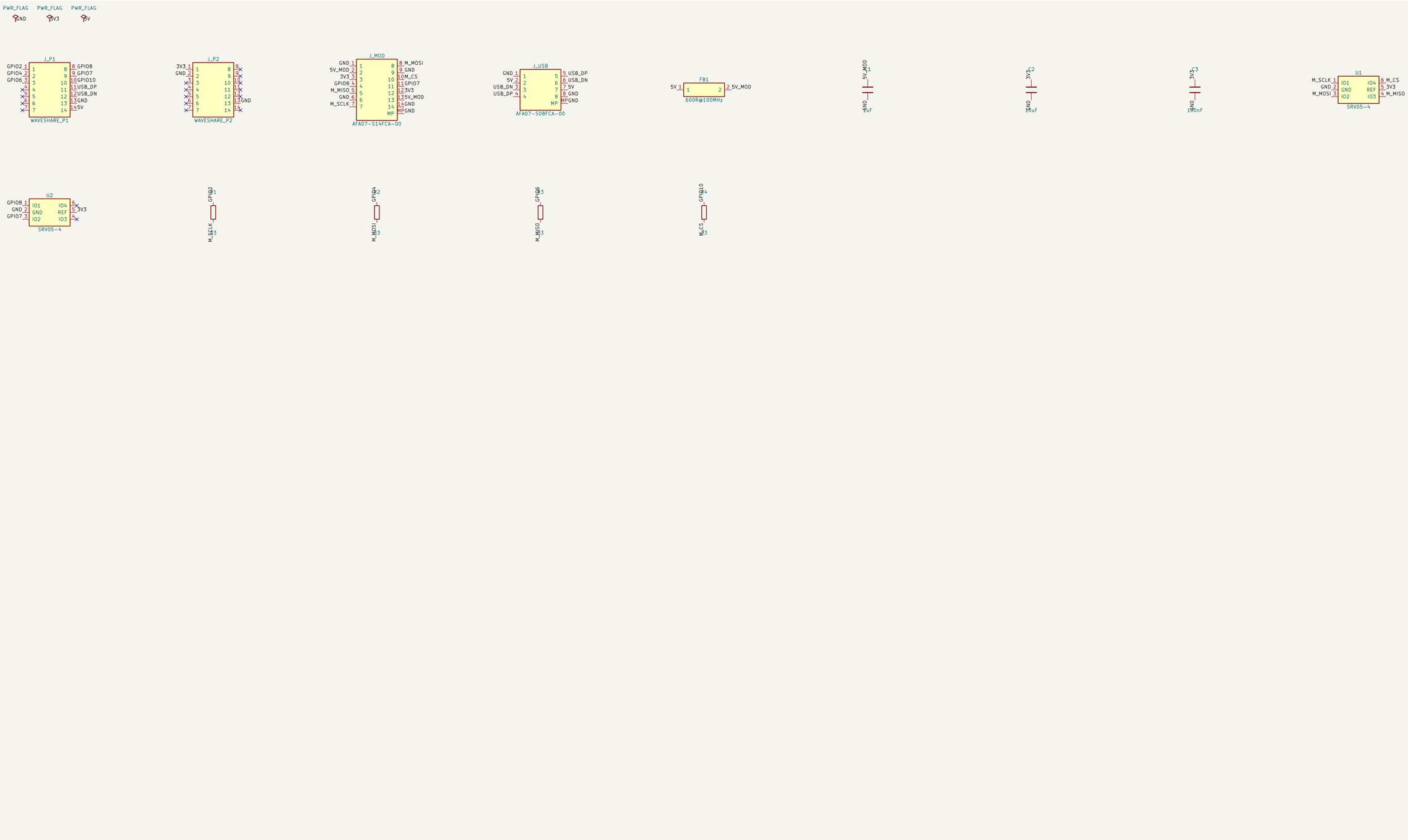


OPad carrier V3: Waveshare headers, USB cable from the USB board, module FFC.

Module FFC: 1 GND | 2 5V | 3 3V3 | 4 ID=GPIO8 | 5 MISO=GPIO6 | 6 GND | 7 SCLK=GPIO2 | 8 MOSI=GPIO4 | 9 GND | 10 CS=GPIO10 | 11 SPARE=GPIO7 | 12 3V3 | 13 5V | 14 GND; a reversed cable maps k to 15−k.

USB FFC: 1 GND | 2 5V | 3 D− | 4 D+ | 5 D+ | 6 D− | 7 5V | 8 GND (the same read from either end): 5V −> P1−14, D+ −> P1−11 (GPIO20), D− −> P1−12 (GPIO19). The USB board filters and protects it.

OPad	
Sheet: /	
File: carrier_v3.kicad_sch	
Title: OPad Controller Carrier V3	
Size: A2	Date: 2026-10-28
KiCad E.D.A. 10.0.7	**Rev: V3.0**
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